HW 10.4. Non-Mankiw Problems

Principles of Microeconomics

October 24, 2025

1. Supply for plastic toys is given by $Q_S = 50P$, and demand for plastic toys is given by $Q_D = 200 - 5P$. Manufacturing the plastic for each toy generates \$2 worth of waste per toy.

   (a) What quantity of toys is sold at the market equilibrium?

   (b) What quantity of toys is sold at the efficient equilibrium?

   (c) Is there deadweight loss at the market equilibrium? Why or why not?

2. Supply for solar cells is given by $Q_S = 4P$, and demand is given by $Q_D = 14 - 2P$. Because they reduce carbon emissions, every solar cell sold generates \$2 worth of external value.

   (a) What quantity of toys is sold at the market equilibrium?

   (b) What quantity of solar cells is sold at the efficient equilibrium?

   (c) What government policy would generate the efficient equilibrium?

3. Tyson is a poultry processing company. Raising and slaughtering a large number of chickens creates significant waste that smells bad and pollutes nearby waterways. Suppose it costs Tyson \$50,000 to deal with the waste in a way that doesn't affect its neighbors, and suppose its neighbors value fresh air and clean water at \$100,000. Explain what happens in each of the following situations. Think about the Coase theorem.

   (a) Tyson has the right to generate smelly, water polluting waste. Negotiation between Tyson and its neighbors is free.

   (b) Generating smelly, water-polluting waste is illegal. Negotiation is free.

   (c) Tyson has the right to generate smelly, water polluting waste. It costs each party \$10,000 to negotiate due to attorney fees.